📊 Credit Risk & Loan Default Analysis (SQL)

📌 Project Overview

This project demonstrates how SQL can be used to analyze loan applications, assess credit risk, and generate business insights that support lending decisions.

The analysis focuses on identifying factors influencing loan approval, evaluating customer creditworthiness, and discovering trends that financial institutions can use to reduce default risk while improving loan portfolio quality.

This project forms part of my Data Analytics portfolio and showcases practical SQL skills for real-world financial data analysis.

🎯 Business Problem

Financial institutions receive thousands of loan applications, making it difficult to quickly identify high-risk applicants.

The objective of this project is to use SQL to:

- Analyze loan approval patterns
- Assess customer credit risk
- Identify characteristics of approved and rejected loans
- Evaluate the relationship between income, employment, experience, and credit score
- Support better lending decisions through data-driven insights

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📂 Dataset

The dataset contains **5,000 loan applications** with customer demographic and financial information.

Key Fields

- Customer ID
- Age
- Gender
- City
- Annual Income
- Employment Type

- Years of Experience
- Credit Score
- Loan Amount
- Loan Approval Status

🛠 Tools Used

- **SQL** (Data Extraction & Analysis)
- Microsoft SQL Server
- GitHub

📈 SQL Analysis Performed

The project includes SQL queries for:

- Total Loan Requests
- Total Loans Approved
- Total Loans Rejected
- Approval Rate Calculation
- Credit Score Analysis
- Loan Approval by Age
- Loan Approval by Employment Type
- Loan Approval by City
- Loan Approval by Years of Experience
- Income vs Loan Amount Analysis
- Customer Demographics
- Aggregate Functions
- GROUP BY Analysis
- Filtering and Sorting
- CASE Statements
- Ranking Queries

📊 Key Insights

Loan Approval

- Total Loan Applications: **5,000**
- Loans Approved: **1,151**
- Loans Rejected: **3,849**
- Overall Approval Rate: **23%**

Credit Score

Applicants with higher credit scores were significantly more likely to receive loan approval, highlighting credit score as one of the strongest predictors of lending decisions.

Employment Status

Self-employed and salaried applicants received the majority of loan approvals, while unemployed applicants experienced substantially lower approval rates.

Experience

Applicants with greater years of professional experience generally showed higher approval rates, suggesting employment stability positively influences lending decisions.

Geographic Distribution

Loan requests were relatively balanced across all cities, indicating consistent market demand across locations.

Income Analysis

Higher-income applicants generally qualified for larger loan amounts and demonstrated stronger approval outcomes.

💼 Business Recommendations

- Prioritize applicants with strong credit scores during initial screening.
- Consider employment stability as a major lending criterion.
- Introduce additional verification procedures for applicants with lower credit scores.
- Develop tailored lending products for different income groups.
- Monitor city-level lending performance to identify emerging credit risks.

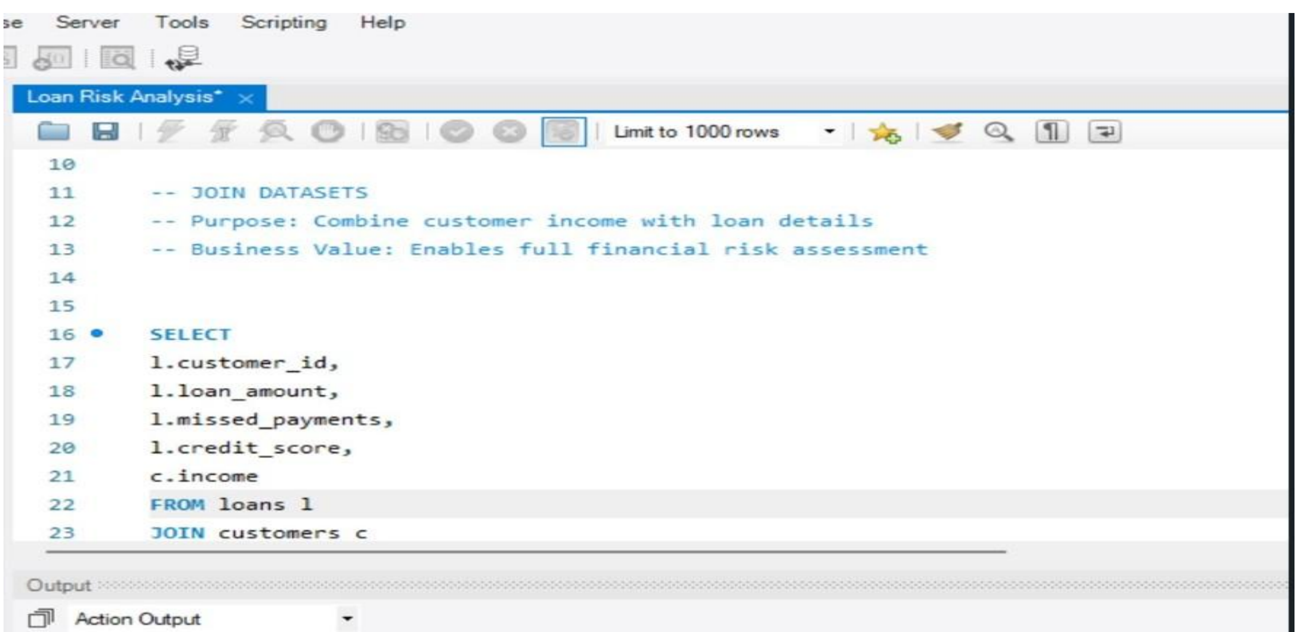
📁 Repository Structure

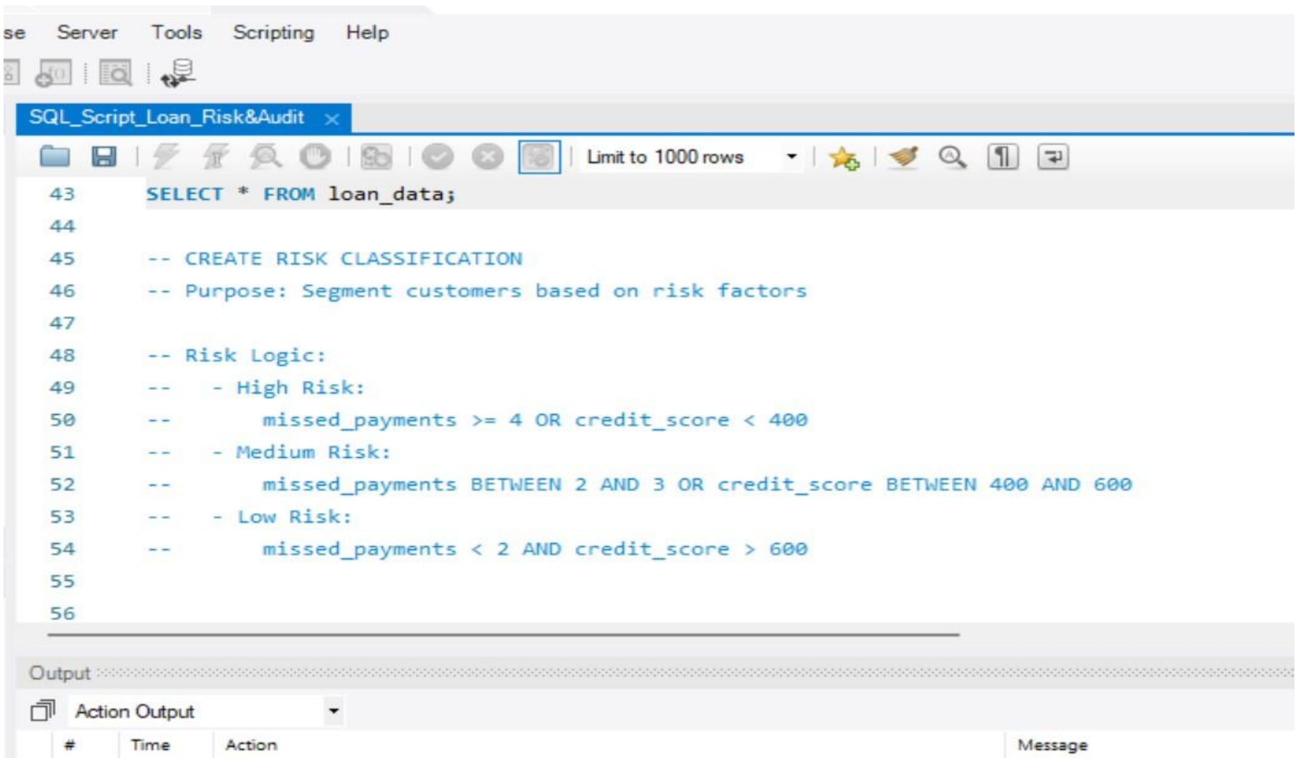
```
Loan-Risk-Audit-Analysis-SQL/
│
├── Dataset/
├── SQL Queries/
├── Results/
├── Documentation/
└── README.md
```

🚀 Skills Demonstrated

- SQL Query Writing
- Data Cleaning
- Data Exploration
- Aggregate Functions
- GROUP BY
- CASE Statements
- Common Table Expressions (CTEs)
- Joins
- Business Intelligence
- Financial Data Analysis
- Credit Risk Assessment
- Data Storytelling

📷 Project Preview





🧑‍💻 About Me

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Data Analyst with hands-on experience in SQL, Python, Microsoft Power BI, Microsoft Excel, financial and business intelligence reporting. Passionate about transforming raw data into actionable insights that support strategic business decisions.

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⭐ If you found this project interesting, feel free to star the repository or connect with me on LinkedIn.
